Lecture 05: Instrumental Variables

EEFE 530

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Intro

Roadmap

1. Motivation

- Endogeneity and omitted variable bias
- Simultaneity and measurement error
- Why selection-on-observables may fail

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2. Instrumental Variables Framework

- Structural vs. reduced form
- Wald estimator (binary instrument)
- 2SLS with multiple instruments

3. Identification Assumptions

- Relevance
- Exclusion restriction
- Independence
- Monotonicity (for LATE)

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4. Interpretation

- LATE and compliers
- Comparison to ATE / ATT
- Policy interpretation

5. Implementation in R

- First stage diagnostics
- Weak instrument tests
- Clustered SEs
- `fixest::feols()` IV syntax

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6. Applications & Pitfalls

- Weak instruments
- Overidentification tests
- Threats to exclusion
- External validity

Learning Objectives

By the end of this lecture, you should be able to:

- Diagnose sources of endogeneity in an applied setting
- Derive the Wald estimator and interpret 2SLS as a ratio of covariances
- State and evaluate the IV identification assumptions
- Explain LATE and identify the population being estimated
- Implement IV estimation in R and interpret first-stage diagnostics
- Assess instrument strength and common specification threats

Today

Instrumental variables (and two-stage least squares)

Upcoming

- Problem Set 4 - due 3/3

Reading

Shift-Share/Bartik instruments

- Mixtape Ch. 7.8.3
- Autor, D. H., Dorn, D., & Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. American economic review, 103(6), 2121-2168.
- Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. American Economic Review, 110(8), 2586-2624.
- Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-experimental shift-share research designs. The Review of Economic Studies, 89(1), 181-213.

Research designs

Selection on observables and/or unobservables

We've been focusing on ***selection-on-observables* designs**, *i.e.*,

$$(\mathbf{Y}_{0i}, \mathbf{Y}_{1i}) \perp\!\!\!\perp \mathbf{D}_i | \mathbf{X}_i$$

for **observable** variables $\mathbf{X}_i$.

***Selection-on-unobservable* designs** replace this assumption with two new (but related) assumptions

1. $(\mathbf{Y}_{0i}, \mathbf{Y}_{1i}) \perp \mathbf{Z}_i$ - There is a variable Z_i that is independent of the potential outcomes.
2. $\mathbf{Cov}(\mathbf{Z}_i, \mathbf{D}_i) \neq \mathbf{0}$ - That variable Z_i is correlated with treatment (D_i).

Selection on observables and/or unobservables

The goal of applied econometrics is to estimate causal parameters. We need to isolate exogenous variation in $\mathbf{D}_i$.

(We want to avoid selection bias.)

- **Selection-on-observables designs** assume that we can control for all *endogenous variation* (selection) in $\mathbf{D}_i$ through a known (and observed) $\mathbf{X}_i$.
- **Selection-on-unobservables designs** assume we can isolate *exogenous variation* in $\mathbf{D}_i$ (generally using some $\mathbf{Z}_i$) and then use this component of $\mathbf{D}_i$ to estimate the causal effect of $\mathbf{D}_i$ on $\mathbf{Y}_i$.

We ignore or remove the *endogenous variation* in $\mathbf{D}_i$ (it's bad).

Which is better?

Both ***selection-on-unobservable*** and ***selection-on-unobservable*** designs represent *identification strategies*. Is one intrinsically prefered to the other?

Both rely on assumptions. The question is which assumptions are defensible in a given application.

1. It's easy to violate the assumptions of either design.

2. **Selection on observables** assumes we know *everything* about selection into treatment—we can remove all of the selection bias (recall our omitted variable bias formula) Difficult in observational settings, and difficult to validate.

3. **Selection on unobservables** assumes we can isolate *some* exogenous variation in $\mathbf{D}_i$, which we then use to estimate the effect of $\mathbf{D}_i$ on $\mathbf{Y}_i$.
   Can defend in some settings, and possible to validate. May have power issues.

Instrumental variables

Introduction

Instrumental variables: (IV).super[.dblue[1]] is the canonical selection-on-unobservables design. We attempt to isolate **exogneous variation** in $\mathbf{D}_i$ via an instrument $\mathbf{Z}_i$.

Note: For the moment, we're lumping together IV and two-stage least squares (2SLS) together—as many people do—even though they are technically different.

Consider some model (structural equation)

$$\mathbf{Y}_i = \beta_0 + \beta_1 \mathbf{D}_i + \varepsilon_i \tag{1}$$

To guarantee consistent (unbiased) OLS estimates for β_1 we need $\mathbf{Cov}(\mathbf{D}_i,\ \varepsilon_i) = \mathbf{0}$.

We cannot test this, so really we need to *assume* $\mathbf{Cov}(\mathbf{D}_i,\ \varepsilon_i) = \mathbf{0}$.
In general, this is a heroic assumption.

Alternative: Estimate β_1 via instrumental variables.

Definition

For our model

$$\mathbf{Y}_i = \beta_0 + \beta_1 \mathbf{D}_i + \varepsilon_i \tag{1}$$

A valid **instrument** is a variable $\mathbf{Z}_i$ such that

1. $\mathbf{Cov}(\mathbf{Z}_i,\ \mathbf{D}_i) \neq \mathbf{0}$
   - our instrument is correlated with treatment
2. $\mathbf{Cov}(\mathbf{Z}_i,\ \varepsilon_i) = \mathbf{0}$
   - and our instrument is uncorrelated with determinants of $\mathbf{Y}_i$ other than D_i **(exclusion restriction)**
   - Another way of stating the exclusion restriction is that the only way that Z_i affects Y_i is through D_i.

Example

Back to the returns to a college degree,

$$\textbf{Income}_i = \beta_0 + \beta_1 \textbf{Grad}_i + \varepsilon_i$$

OLS is likely biased.

Why?

What if that state conducts a random **lottery** for scholarships?

Let $\textbf{Lottery}_i$ denote an indicator for whether $\boldsymbol{i}$ won a lottery scholarship.

Note: It's important to consider who is eligible and/or applied to the lottery.

1. $\textbf{Cov}(\textbf{Lottery}_i,\ \textbf{Grad}_i) \neq \mathbf{0}$ if scholarships increase graduation rates.
2. $\textbf{Cov}(\textbf{Lottery}_i,\ \varepsilon_i) = \mathbf{0}$ since the lottery is randomized.

Instrumental variables

The IV estimator

The IV estimator for our model

$$\text{Y}_i = \beta_0 + \beta_1 \text{D}_i + \varepsilon_i \tag{1}$$

with (valid) instrument $\mathbf{Z}_i$ is

$$\hat{\beta}_{\text{IV}} = \left(\text{Z}'\text{D}\right)^{-1} \left(\text{Z}'\text{Y}\right)$$

–

If you have no covariates, then

$$\hat{\beta}_{\text{IV}} = \frac{\text{Cov}(\text{Z}_i,\ \text{Y}_i)}{\text{Cov}(\text{Z}_i,\ \text{D}_i)}$$

If you have additional (exogenous) covariates $\mathbf{X}_i$, then

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Z}_i & \mathbf{X}_i \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} \mathbf{D}_i & \mathbf{X}_i \end{bmatrix}$$

Two-stage least squares

Setup

We often implement IV as a two-stage process known as **two-stage least squares** (2SLS).

First stage: Estimate the effect of the instrument $\mathbf{Z}_i$ on our endogenous treatment variable $\mathbf{D}_i$ and (predetermined) covariates $\mathbf{X}_i$. Save $\widehat{\mathbf{D}}_i$.

$$\mathbf{D}_i = \gamma_1 \mathbf{Z}_i + \gamma_2 \mathbf{X}_i + u_i$$

Second stage: Estimate the original structural model replace $\mathbf{D}_i$ with $\widehat{\mathbf{D}}_i$.

$$\mathbf{Y}_i = \beta_1 \widehat{\mathbf{D}}_i + \beta_2 \mathbf{X}_i + \varepsilon_i$$

Note: The controls $\mathbf{X}_i$ must match in the first and second stages.

IV estimation

This two-step procedure, with a valid instrument, produces an estimator $\hat{\beta}_1$ that is consistent for β_1.

$$\hat{\beta}_{2SLS} = \left(D' P_Z D\right)^{-1} \left(D' P_Z Y\right)$$
$$P_Z = Z(Z'Z)^{-1} Z'$$

where $\mathbf{D}$ is a matrix of our treatment and predetermined covariates $(\mathbf{X}_i)$ and $\mathbf{Z}$ is a matrix of our instrument and our predetermined covariates.

IV estimation

Important notes

- The controls $(\mathbf{X}_i)$ must match in the first and second stages.
- If you have exactly **one instrument** and exactly **one endogenous variable**, then 2SLS and IV are identical.
- Your second-stage standard errors are not correct.

The reduced form

In addition to the regressions within the two stages of 2SLS

1. $\mathbf{D}_i = \gamma_1 \mathbf{Z}_i + \gamma_2 \mathbf{X}_i + u_i$
2. $\mathbf{Y}_i = \beta_1 \widehat{\mathbf{D}}_i + \beta_2 \mathbf{X}_i + \varepsilon_i$

there is a third important and related regression:

the reduced form.

The **reduced form** regresses the outcome $\mathbf{Y}_i$ (LHS of the second stage) on our instrument $\mathbf{Z}_i$ and covariates $\mathbf{X}_i$ (RHS of the first stage).

$$\mathbf{Y}_i = \pi_1 \mathbf{Z}_i + \pi_2 \mathbf{X}_i + u_i$$

Thus, the reduced form provides a consistent estimate of the causal effect of our instrument on the outcome.

The reduced form

While the reduced form estimates the causal effect of the instrument on our outcome, we're usually interested in the effect of *treatment* $(\mathbf{D}_i)$.

That said, the reduced form is still incredibly helpful/important:

- Clarifies your source of identifying variation.
- Does not suffer from *weak instruments* problems.
- Only requires $\mathbf{Cov}(\mathbf{Z}_i,\ \varepsilon_i) = \mathbf{0}$.
- Offers insights into your estimates

$$\widehat{\beta}_1^{\text{2SLS}} = \frac{\widehat{\pi}_1}{\widehat{\gamma}_1}$$

when you have exactly one instrument.

The reduced form, intuition

This expression for the 2SLS (and IV) estimator can be very helpful.

$$\widehat{\beta}_1^{\text{2SLS}} = \frac{\widehat{\pi}_1}{\widehat{\gamma}_1} = \frac{\textbf{Reduced-form estimate}}{\textbf{First-stage estimate}}$$

What's the interpretation/intuition?

Back to our example: $\widehat{\beta}_1 =$ estimated effect of college graduation on income.

$\widehat{\pi}_1$ gives the estimated causal effect of the scholarship lottery on income.

We need to consider how winning the lottery affects graduation. We need to rescale $\widehat{\pi}_1$ if the effect of lottery on graduation is $< 100\%$.

$\widehat{\gamma}_1$ estimates the effect of winning the scholarship lottery on graduation — the share of winners who graduated *due* to winning.

We can scale with $\widehat{\gamma}_1$!

The reduced form, example

Let's gain some insight into scaling the reduced form. Imagine that 50% of lottery winners graduate from college due to the lottery, *i.e.*, $\widehat{\gamma}_1 = 0.50$.

Imagine none of the applicants would have graduated otherwise

Our reduced-form estimate of $\widehat{\pi}_1 = \mathbf{\$5,000}$ says that lottery winners make \$5,000 more than the control group, on average.

However, half of the winners did not graduate, so $\widehat{\pi}_1$ "underestimates" the effect of college graduation by combining graduates by nongraduates.

Thus, we want to double $\widehat{\pi}_1$, *i.e.*, divide by $\widehat{\gamma}_1$:

$\widehat{\pi}_1/\widehat{\gamma}_1 = \mathbf{\$5,000}/\mathbf{0.50}$ = \$10,000.

Q: How do we get this intuitive expression? $\left(\widehat{\beta}_1^{\text{IV}} = \dfrac{\widehat{\pi}_1}{\widehat{\gamma}_1}\right)$

Derivation

$$\widehat{\beta}_1^{\text{IV}} = (\text{Z}'\text{D})^{-1} (\text{Z}'\text{Y})$$

$$= \left(\widetilde{\text{Z}}'\widetilde{\text{D}}\right)^{-1} \left(\widetilde{\text{Z}}'\text{Y}\right)$$ applying FWL to reduce $\mathbf{D}$ and $\mathbf{Z}$ to vectors.

$$= \frac{\text{Cov}\left(\widetilde{\text{Z}}_i,\ \text{Y}_i\right)}{\text{Cov}\left(\widetilde{\text{Z}}_i,\ \widetilde{\text{D}}_i\right)}$$

$$= \frac{\text{Cov}\left(\widetilde{\text{Z}}_i,\ \text{Y}_i\right) / \text{Var}\left(\widetilde{\text{Z}}_i\right)}{\text{Cov}\left(\widetilde{\text{Z}}_i,\ \widetilde{\text{D}}_i\right) / \text{Var}\left(\widetilde{\text{Z}}_i\right)}$$

$$= \frac{\widehat{\pi}_1}{\widehat{\gamma}_1} \quad \checkmark$$

Let's push a bit deeper into IV's mechanics and intuition.

IV: Mechanics and intuition

Setup

In this section, we'll use medical trials as a working example.

We are interested in the regression model for the effect of some treatment (*e.g.*, blood-pressure medication) on medical outcome $\mathbf{Y}_i$

$$\mathbf{Y}_i = \beta_0 + \beta_1 \mathbf{D}_i + \varepsilon_i$$

$\mathbf{D}_i$ indicates whether $\boldsymbol{i}$ *takes* the treatment (medication). $\boldsymbol{\varepsilon}_i$ captures all other factors that affect $\mathbf{Y}_i$.

Or in potential-outcomes framework:

$$\begin{aligned} \mathbf{Y}_i &= \mathbf{Y}_{1i}\mathbf{D}_i + \mathbf{Y}_{0i}(1 - \mathbf{D}_i) \\ \mathbf{Y}_{0i} &= \beta_0 + \varepsilon_i \\ \mathbf{Y}_{1i} &= \mathbf{Y}_{0i} + \beta_1 \end{aligned}$$

Credit/thanks go to Michael Anderson via Ed Rubin for this example—and much of these notes.

Research design

Goal: *Estimate the effect of blood-pressure medication* on blood pressure.

**Challenge: *Selection bias:* Even if treatment reduces blood pressure, selection bias fights against the estimated effect.

Solution: *Randomized medical trial:* Ask randomly chosen individuals in treatment group to take the pill. Control individual get placebo (or nothing).

Analysis 1: *Intention to treat (ITT):* $\widehat{\beta}_1^{\text{ITT}} = \overline{\text{Y}}_{\text{Trt}} - \overline{\text{Y}}_{\text{Ctrl}}$

ITT problem: *Bias from noncompliance:* People don't always follow rules.
E.g., treated folks who don't take pills; control folks who take pills.

Analysis 2: *IV!*

Instrument medication D_i with intention to treat Z_i.

The IV solution

First question: Is $\mathbf{Z}_i$ a valid instrument for $\mathbf{D}_i$?

1. $\mathbf{Cov}(\mathbf{Z}_i,\ \varepsilon_i) = \mathbf{0}$ as $\mathbf{Z}_i$ was randomly assigned (exclusion restriction).
2. $\mathbf{Cov}(\mathbf{Z}_i,\ \mathbf{D}_i) \neq \mathbf{0}$ if assignment to treatment changes the likelihood you take the pills (first stage).

$\therefore \mathbf{Z}_i$ is a valid instrument for $\mathbf{D}_i$ and IV consistently estimates $\beta_\mathbf{1}$.

Noncompliance

Noncompliant individuals do not abide by their treatment assignment.

Let's see how IV "solves" this problems.

First, assume noncompliance only affects treated individuals—*i.e.*, treated folks sometimes don't take their pills; control folks never take pills.

This is often called one-sided noncompliance. This would be the case of the COVID vaccine trials where the vaccine is not publicly available so subjects in the control group **cannot** take get the shot.

Noncompliance, continued

The **first stage** recovers the share of treatment individuals who take the pill

$$\mathbf{D}_i = \gamma_1 \mathbf{Z}_i + u_i$$

i.e., if 50% of treated individuals take the medication, $\widehat{\gamma} = 0.50$.

The **reduced form** estimates the *ITT*

$$\mathbf{Y}_i = \pi_1 \mathbf{Z}_i + v_i$$

which we know IV rescales using the first stage

$$\widehat{\beta}_1^{\text{IV}} = \frac{\widehat{\pi}_1}{\widehat{\gamma}_1} = \frac{\widehat{\pi}_1}{0.50} = 2 \times \widehat{\pi}_1$$

Noncompliance, continued

IV solves the noncompliance issue by rescaling by the rate of compliance.

If everyone perfectly complies, then $\widehat{\gamma}_1 = \mathbf{1}$ and $\widehat{\beta}_1^{\text{IV}} = \widehat{\pi}_1/\mathbf{1} = \widehat{\beta}_1^{\text{ITT}}$.

Further example: N_{Trt} = 10; trt. compliance = 50%; ctrl. compliance = 100%.

$$\overline{\mathbf{Y}}_{\text{Trt}} = \frac{5(\beta_0 + \beta_1) + 5(\beta_0)}{10} = \beta_0 + \frac{\beta_1}{2}$$

and $\overline{\mathbf{Y}}_{\text{Ctrl}} = \beta_0$.

So our reduced-form estimate (the ITT) is $\widehat{\gamma}_1 = \dfrac{\beta_1}{2}$ (half the true effect).

IV consistently estimates β_1 via rescaling the ITT by the rate of compliance

$$\widehat{\beta}_1^{\mathrm{IV}} = \frac{\pi}{\gamma} = \frac{\beta_1/2}{1/2} = \beta_1$$

Example

This can have big implications in many settings including medical trials with imperfect compliance.

Doctors and public health officials advocate for cancer screening as cost-effective medical intervention.

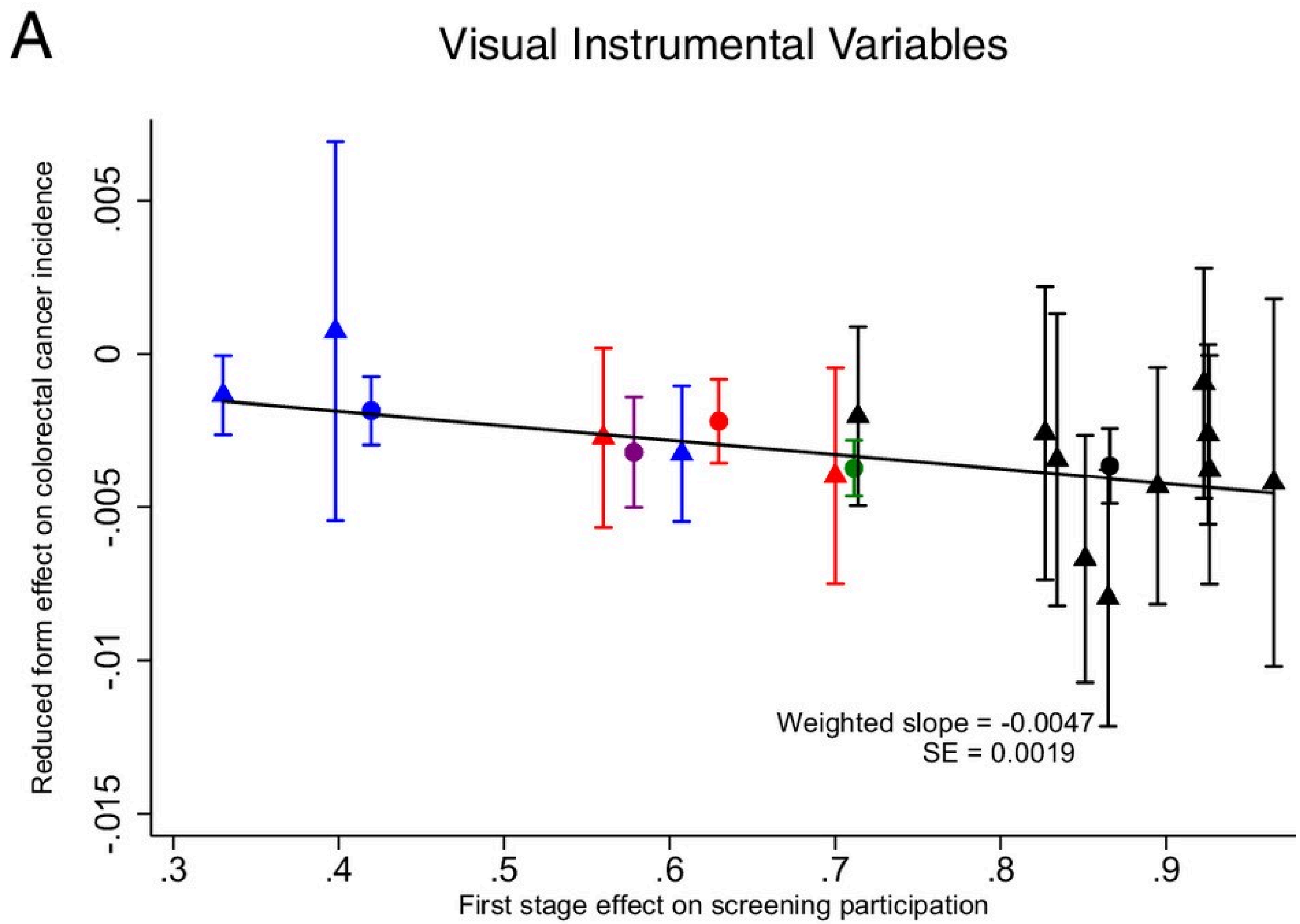
However, randomized trials for that invite participants to free colonoscopy and sigmoidoscopy show very different results.

So do screenings work?

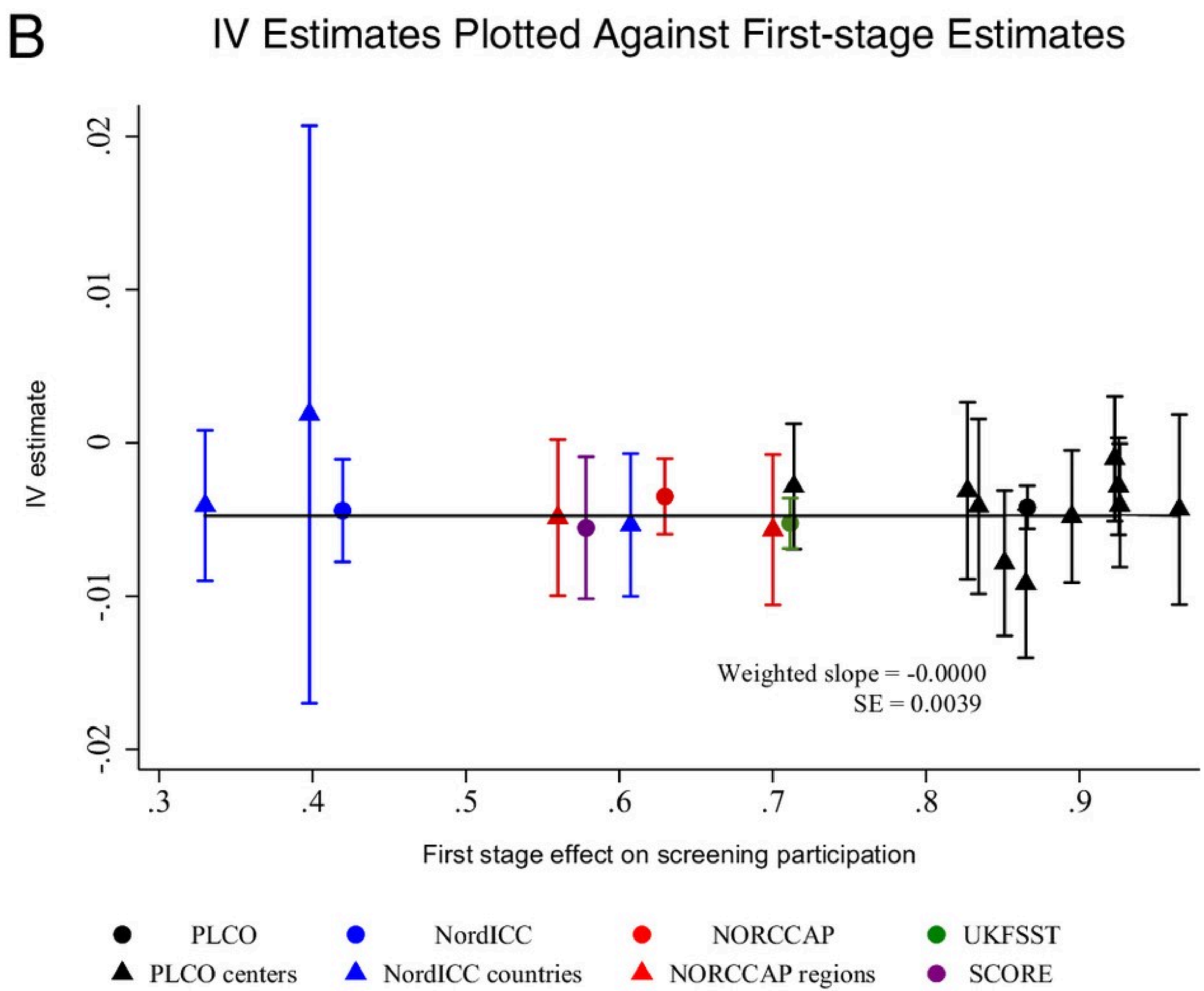
Enter IV!

Angrist and Hull (2023)

Example


A
Visual Instrumental Variables
Reduced form effect on colorectal cancer incidence
.005
0
-.005
-.01
-.015
.3
.4
.5
.6
.7
.8
.9
First stage effect on screening participation
Weighted slope = -0.0047
SE = 0.0019

Example


B
IV Estimates Plotted Against First-stage Estimates
IV estimate
.02
.01
0
-.01
-.02
.3
.4
.5
.6
.7
.8
.9
First stage effect on screening participation
Weighted slope = -0.0000
SE = 0.0039
PLCO
NordICC
NORCCAP
UKFSST
PLCO centers
NordICC countries
NORCCAP regions
SCORE

Takeaways

Main points

1. IV **rescales** the causal effect of $\mathbf{Z}_i$ on $\mathbf{Y}_i$ by the causal effect of $\mathbf{Z}_i$ on $\mathbf{D}_i$.

2. IV **does not** compare treated compliers to untreated non-compliers.

Such a comparison/estimator would re-introduce selection bias.

Thus far, we assumed homogeneous treatment effects.

Q: What happens .*when treatment effects are heterogeneous*?

IV + heterogeneity

Recap

So far we've discussed two concepts of treatment effects (really 3 if you count ITT).

1. **Average treatment effect (ATE):** The average treatment effect for an individual randomly drawn from our sample.
2. **Average Treatment effect on the treated (ATT]):** The average treatment effect for a *treated* individual randomly drawn from our sample.

When we assume homogeneous/constant treatment effects, ATE = ATT

Q: If treatment effects vary, then what do IV and 2SLS estimate?

A: Not ATE.

And not ATT.

They estimate the LATE.

See Angrist, Imbens, and Rubin (1996).

The LATE

IV generally estimates the **LATE** — the Local Average Treatment Effect.

Recall: IV works by isolating variation in $\mathbf{D}_i$ induced by our instrument $\mathbf{Z}_i$.

In other words: IV focuses on the individuals whose $\mathbf{D}_i$ changes **due** to $\mathbf{Z}_i$.

Angrist, Imbens, and Rubin (1996) call these individuals **compliers**.

However, *compliers* are only one of four possible groups.

1. **Compliers** $\mathrm{D}_i = 1$ iff $\mathrm{Z}_i = 1$.
2. **Always-takers** $\mathrm{D}_i = 1 \; \forall \; \mathrm{Z}_i$.
3. **Never-takers** $\mathrm{D}_i = 0 \; \forall \; \mathrm{Z}_i$.
4. **Defiers** $\mathrm{D}_i = 1$ iff $\mathrm{Z}_i = 0$.

1. **Compliers** Only take pills **when treated**.
2. **Always-takers Always** take pills.
3. **Never-takers Never** take pills.
4. **Defiers**
   Only take pills **when untreated**.

The LATE

Because IV only uses variation in $\mathbf{D}_i$ that correlates with $\mathbf{Z}_i$, IV mechanically drops *always-takers* and *never-takers*.

Most IV derivations/applications assume away the existence of *defiers*.

What's the problem with defiers and what assumption does IV usually make to deal with them?

We'll come back to this later.

Thus, IV estimates a treatment effect **using only *compliers***.

Hence the "local" in *local average treatment effect*.

We can only make inference on a subset of the population of interest - **those who are induced into treatment by the instrument.**

The LATE: Medical-trial example

Imagine treatment works for some $(\beta_{1,i} < 0)$ and not for others $(\beta_{1,j} = 0)$.

Suppose individuals know their response to blood-pressure medication.

- $\beta_{1,i} < 0$ individuals always take the pill.
- $\beta_{1,j} = 0$ individuals only take the pill when treated.

Then our compliers will be individuals for whom $\beta_{1,j} = 0$.

Thus, IV's LATE will indicate no treatment effect $\left(\widehat{\beta}_1^{\text{IV}} = 0\right)$ even though the true ATE is nonzero $\left(\beta_1^{\text{ATE}} \neq 0\right)$.

The LATE

Q: So is IV actually inconsistent?

A: It depends what you are trying to estimate (and how you interpret it).

IV doesn't estimate the ATE or ATT, so it would be inconsistent for them.

IV estimates the *local* average treatment effect.

Takeaway 1: Because IV identifies off of compliers, it estimates an average treatment effect for these individuals (who *comply* with the instrument).

Takeaway 2: Different instruments have different LATEs.

Monotonicity

We've already written down the two classical IV/2SLS assumptions

- **First stage:** $\mathrm{Cov}(\mathrm{Z}_i,\ \mathrm{D}_i) > 0$
- **Exclusion restriction:** $\mathrm{Cov}(\mathrm{Z}_i,\ \varepsilon_i) = 0$

but we need a third assumption to ensure IV's complier-based LATE interpretation.

- **Monotonicity (Uniformity):** $\mathbf{D}_i(z) \geq \mathbf{D}_i(z')$ or $\mathbf{D}_i(z) \leq \mathbf{D}_i(z')\ \ \forall i$
  - $\longrightarrow$ Heckman: *Uniformity* of responses *across persons.*
  - $\longrightarrow$ Imbens and Angrist (1994): Instrument has monotone effect on $\mathbf{D}_i$.

Monotonicity

If “defiers” exist, then monotonicity/uniformity is violated.

In this case, the IV estimand is

$$\frac{\tau_c \Pr(\text{complier}) - \tau_d \Pr(\text{defier})}{\Pr(\text{complier}) - \Pr(\text{defier})}$$

which is not bound between τ_c and τ_d.

Example: $\tau_c = 1$ and $\tau_d = 2$. $\Pr(\text{complier}) =$ 2/3 and $\Pr(\text{defier}) =$ 1/3.

Then the “LATE” is 0.

Until now, we've focused on using a single instrument.

The 2SLS estimator accommodates multiple instruments.

Whether you can find multiple valid instruments is another question.

Multiple instruments

Motivation

Q: Why include multiple instruments?

A: Multiple instruments can capture more variation in $\mathbf{D}_i$ (efficiency).

Using terminology from the *system-of-equations* literature,

- one instrument for one endogenous variable: *just identified*
- multiple instruments for one endogenous variable: *over identified*

In practice

With (valid) instruments $\mathbf{Z_{1i}}$ and $\mathbf{Z_{2i}}$, the first stage becomes

$$\text{D}_i = \gamma_0 + \gamma_1 \text{Z}_{1i} + \gamma_2 \text{Z}_{2i} + \gamma_3 \text{X}_i + u_i$$

while the second stage is still

$$\text{Y}_i = \beta_0 + \beta_1 \widehat{\text{D}}_i + \beta_2 \text{X}_i + v_i$$

Weak Instruments

Recall how the first stage estimate affects the 2SLS estimate.

$$\widehat{\beta}^{\mathbf{2SLS}} = \frac{\widehat{\pi}}{\widehat{\gamma}}$$

where $\widehat{\boldsymbol{\pi}}$ is the reduced form estimate and $\widehat{\gamma}$ is the first stage estimate.

$\widehat{\gamma}$ enters this formula in the **denominator** and therefore has a **nonlinear** effect on $\widehat{\beta}^{\mathbf{2SLS}}$

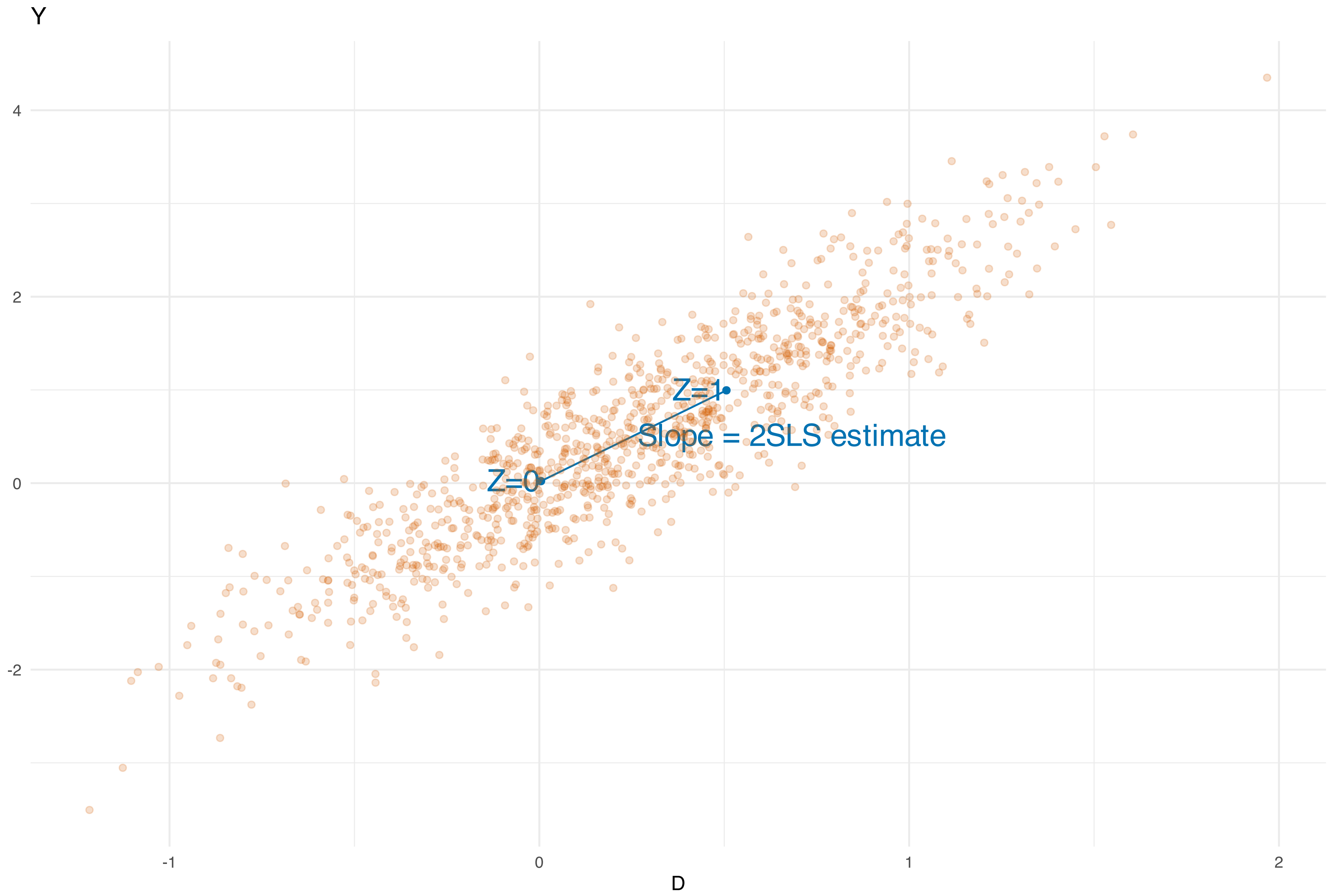
When $\boldsymbol{\gamma}$ is small then small variation in $\widehat{\gamma}$ can have really big impacts on $\widehat{\beta}^{\mathbf{2SLS}}$.

Let's look at this with some simulations

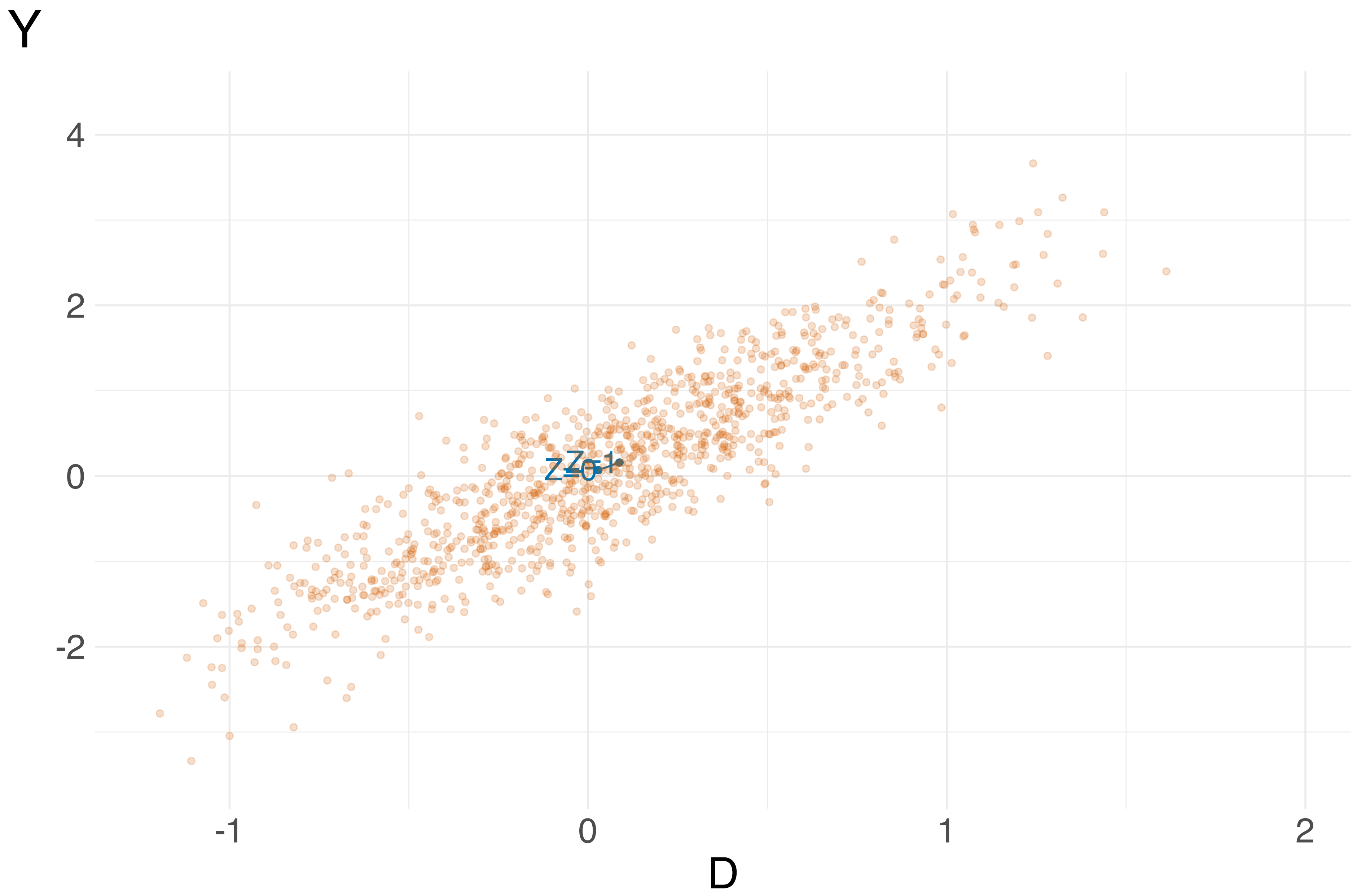
Note: Acknowledgement to Paul Goldsmith-Pinkham for these simulations.

2SLS simulation:

- binary instrument
- First stage coefficient = 0.5
- β = 2
- Variation in 2sLS estimation comes from variation in first stage
- In this setting the length of the interval corresponds to the variation/strength of the instrument

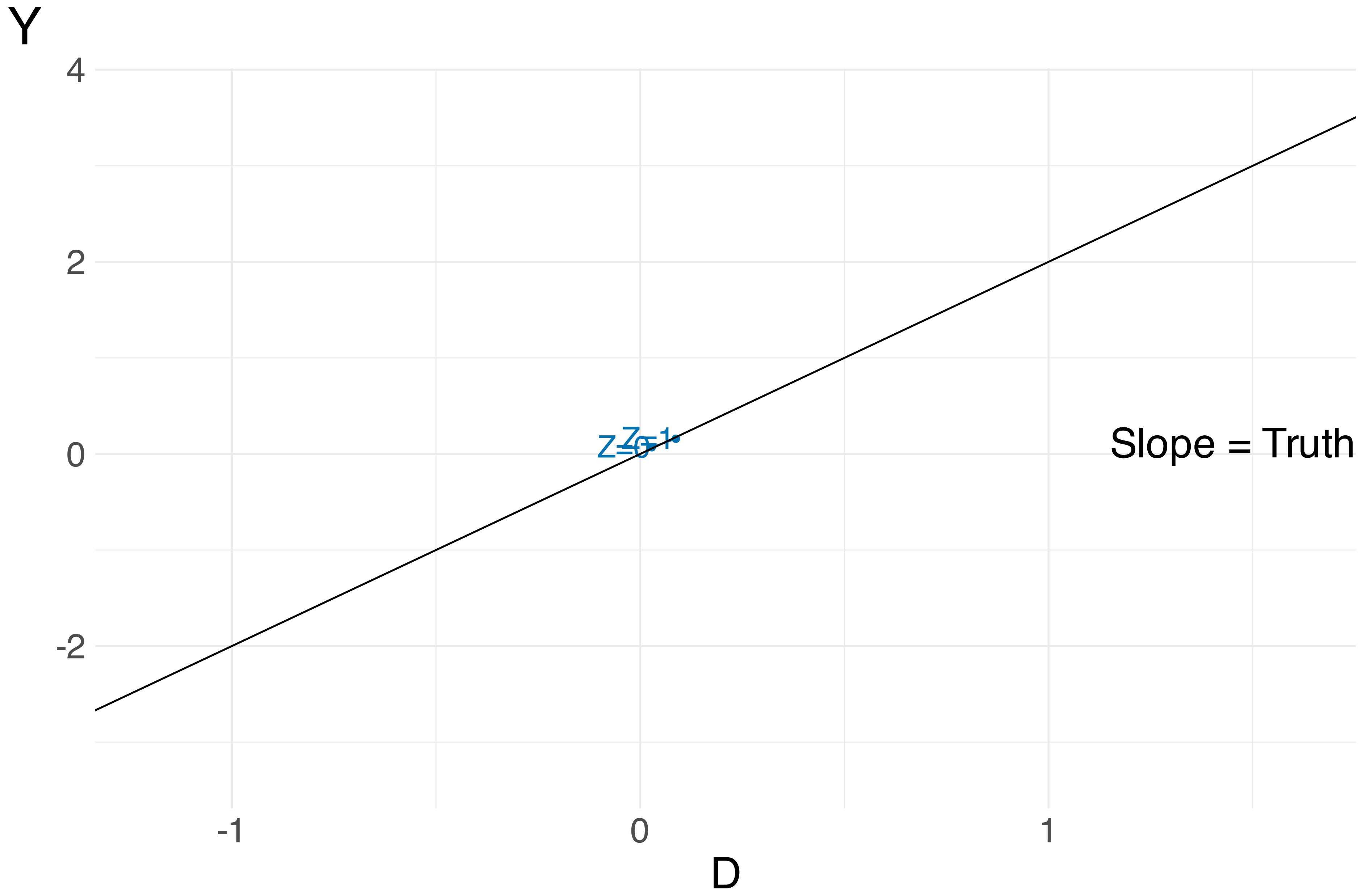

Y
4
2
0
-2
-1
0
1
2
D
Z=1
Z=0
Slope = 2SLS estimate

First stage coefficient = 0.5; β = 2

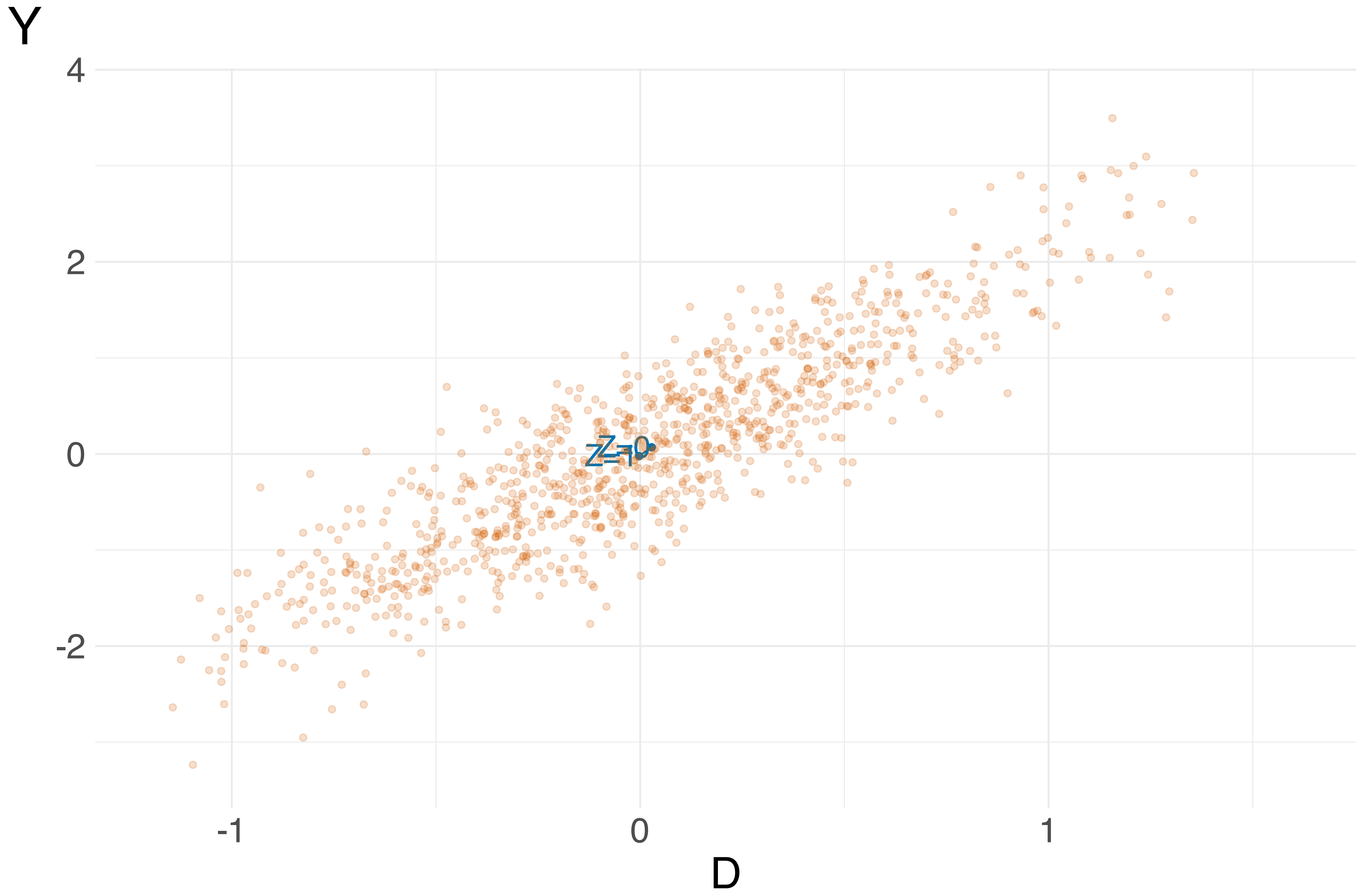


Y
4
2
0
-2
-1
0
1
2
D
Z=1
Z=0

First stage coefficient = 0.1; β = 2


Y
4
2
0
-2
-1
0
1
D
Z=0
Z=1
Slope = Truth

Variation to estimate true effect is tiny.

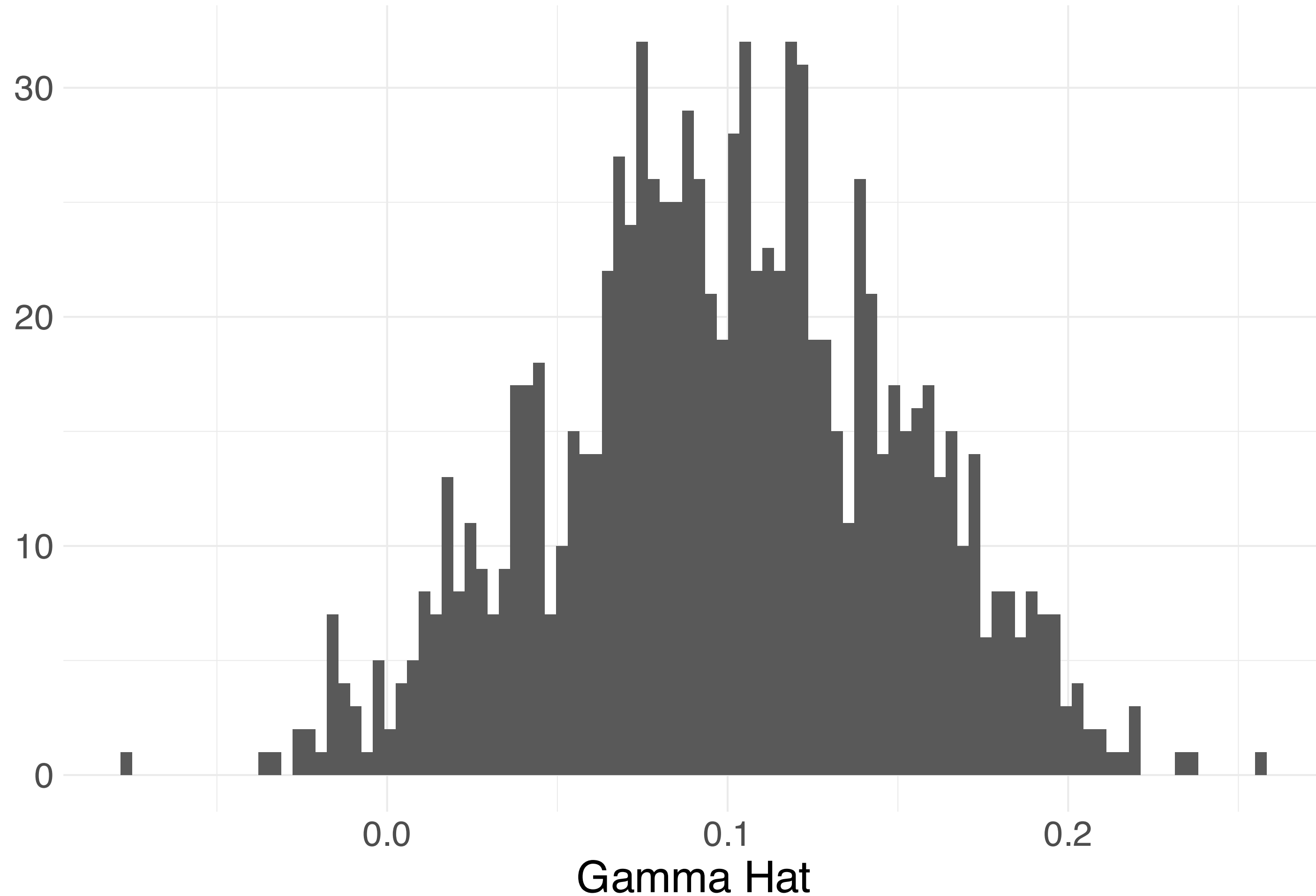

Y
4
2
0
-2
-1
0
1
D
Z=0
Z=1

First stage coefficient = 0.01; β = 2

Let’s look at how this impacts the 2SLS estimates.

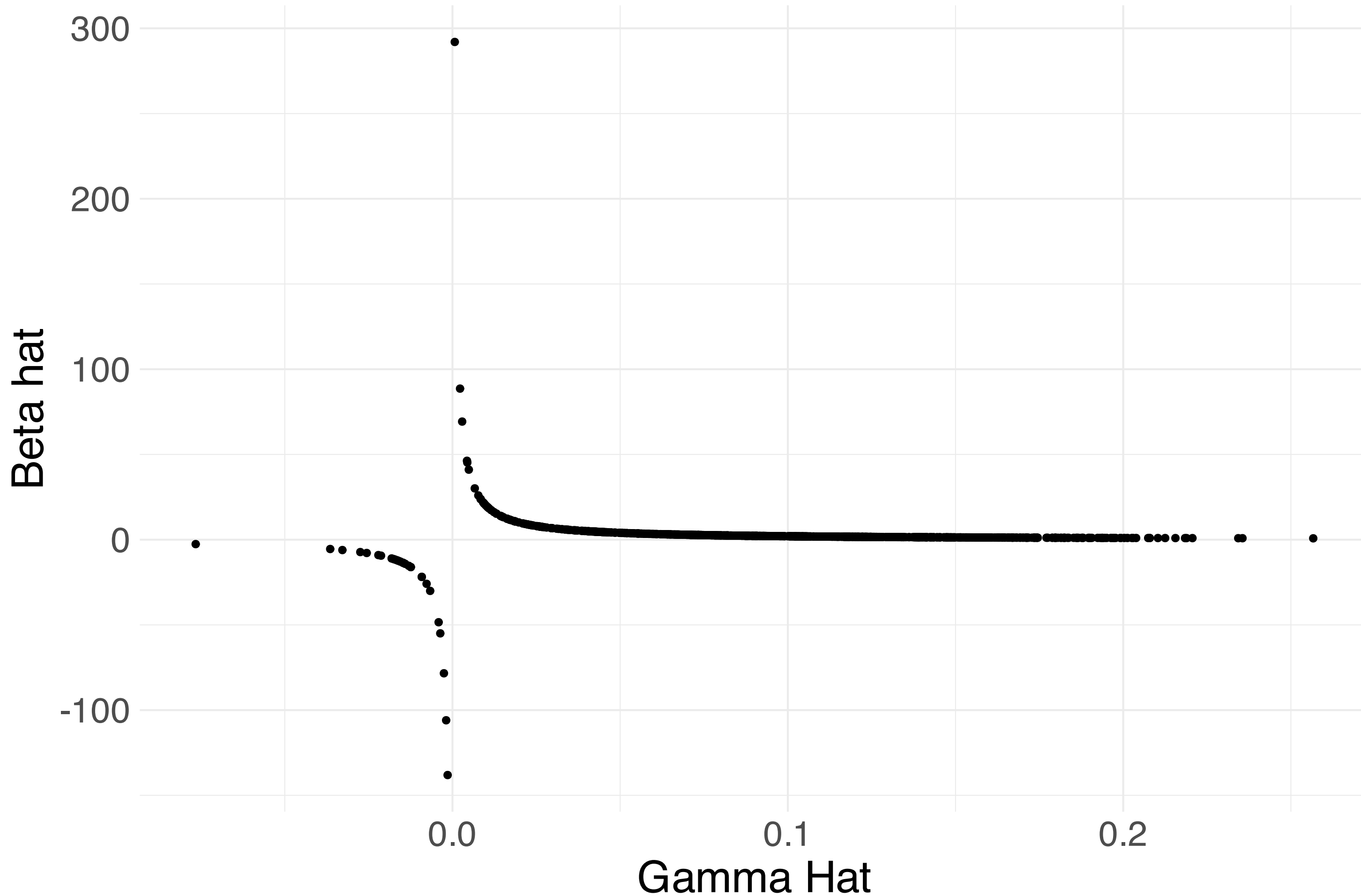
We’ll start with first stage coefficient = 0.01

The first stage estimates look okay


30
20
10
0
0.0
0.1
0.2
Gamma Hat

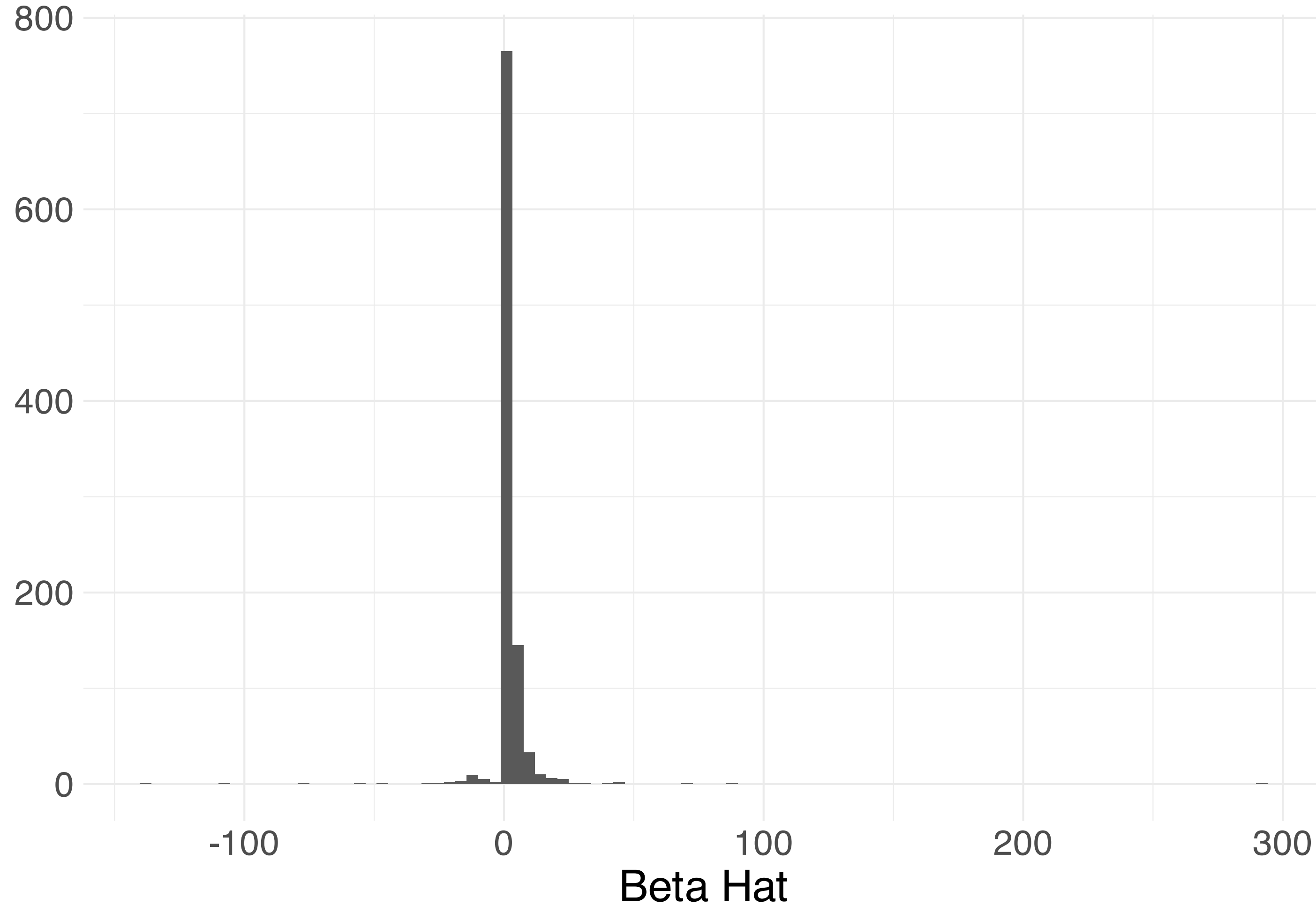
However, the problems show up looking at the relationship between $\widehat{\gamma}$ and $\widehat{\beta}$.

This relationship is nonlinear and undefined when $\widehat{\gamma} = 0$.

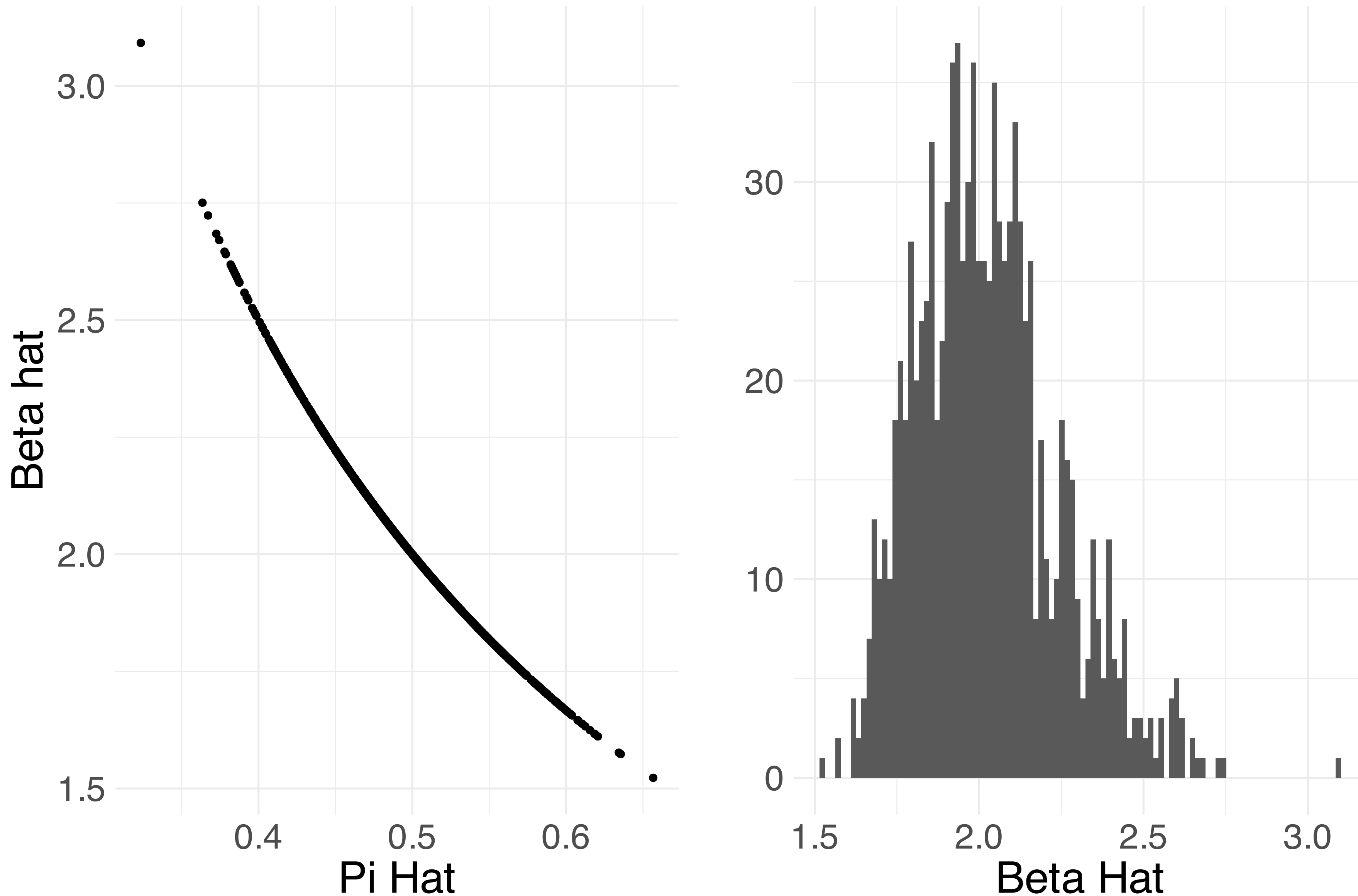

300
200
100
0
-100
Beta hat
0.0
0.1
0.2
Gamma Hat

If the first stage is small and marginally significant (estimate = 0.1, se = 0.05 then a researcher may proceed with the analysis.

However, the distribution for $\widehat{\beta}$ is non normal, and asymptotic normality is a bad approximation.


800
600
400
200
0
-100
0
100
200
300
Beta Hat

This is not the case if the first stage is larger in magnitude (0.5).



Solutions to weak instruments: Pretesting

- One solution is to test whether the first stage relationship is “strong”
- A common approach is to look at the F-statistic
- Staiger and Stock (1997) and Stock and Yogo (2002) develop a rule of thumb of an F-stat above 10
- Bias won’t be larger than 10% with size of 5%
- Key assumption is homoskedasticity

Solutions to weak instruments: Pretesting

- Montieal and Pflueger (2013) have a heteroskedasticity-robust test
  - $\longrightarrow$ cutoff is closer to 23
- **Lee et al. (2022)** show that current practice focuses on the β term as opposed to the t-statistic
- Need a stronger F-stat of over 100!!

Solutions to weak instruments: Pretesting

- Angrist & Kolesar (2023) claim that weak instruments are not a major concern in the just identified case
- Endogeneity needs to **very** high for the bias to outweigh the increase in noise.
- When endogeneity is less that 0.76, CI are distorted by no more than 5% for any F-stat.
- Focus on screening on the *sign* of the first stage as opposed to the F-stat

2SLS and **R**

estimatr

You can implement 2SLS/IV in many ways in .mono[R]. We'll use `estimatr` and `iv_robust()`.

First let's create some sample data.

```
# Set seed
set.seed(12345)
# Sample size
n <- 100
# Define our variance-covariance matrix (D, ε, Z)
sig <- matrix(data = c(1, 0.5, 0.5, 0.5, 1, 0, 0.5, 0, 1), ncol = 3)
# Our vector of means (D, ε, Z)
mu = c(10, 0, 3)
# Draw n observations; convert to tibble
sample_data <- as.data.table(mvrnorm(n = n, mu = mu, Sigma = sig))
# Name variables
names(sample_data) <- c("D", "e", "Z")
# Calculate Y
sample_data[,Y := 7 + 2 * D + e]
```

Specifically, we give `iv_robust()` the relationship that we want separted from the instrument by `|`

, *e.g.*,

```r
# Estimate 2SLS
twosls <- iv_robust(Y ~ D | Z, data = sample_data, se_type = "classical")
tidy(twosls)[1:5]
```

```
#>          term estimate std.error statistic      p.value
#> 1 (Intercept) 5.637525 1.8003223  3.131398 2.293598e-03
#> 2           D 2.121242 0.1842559 11.512480 6.577085e-20
```

Now in two stages!

We can also estimate 2SLS in two stages.

```r
# First stage
stage1 <- lm_robust(D ~ Z, data = sample_data, se_type = "classical")
# First-stage results
tidy(stage1)[1:5]
```

```
#>          term  estimate  std.error statistic      p.value
#> 1 (Intercept) 8.2078914 0.29282475 28.030047 1.354379e-48
#> 2           Z 0.5366599 0.09583812  5.599649 1.962503e-07
```

Second stage

We just need to add $\widehat{\mathrm{D}}_i$ to our dataset, and regress Y on $\widehat{\mathrm{D}}_i$.

```r
# Add fitted (first-stage) values to data
sample_data[,D_hat := stage1$fitted.values]
# Second stage
stage2 <- lm_robust(Y ~ D_hat, data = sample_data, se_type = "classical")
# Second-stage results
tidy(stage2)[1:5]
```

```
#>          term estimate std.error statistic      p.value
#> 1 (Intercept) 5.637525 5.0620048  1.113694 2.681345e-01
#> 2       D_hat 2.121242 0.5180762  4.094460 8.707982e-05
```

LATE

We could also calculate the LATE 'by hand'.

Recall that:

$$\widehat{\beta}^{\mathbf{IV}} = \frac{\widehat{\pi}}{\widehat{\gamma}}$$

where $\widehat{\pi}$ is the reduced form estimate and $\widehat{\gamma}$ is the first stage estimate.

Let's first estimate the reduced form (ITT).

```r
# First stage
reduce <- lm_robust(Y ~ Z, data = sample_data, se_type = "classical")
# First-stage results
tidy(reduce)[1:5]
```

```
#>          term  estimate std.error statistic      p.value
#> 1 (Intercept) 23.048450 0.8494977  27.13186 2.297274e-47
#> 2           Z  1.138386 0.2780307   4.09446 8.707982e-05
```

LATE

From `reduce` $\widehat{\pi}$ =

```r
coef(reduce)[2]
```

```
#>        Z
#> 1.138386
```

and from `stage1` $\widehat{\gamma}$ =

```r
coef(stage1)[2]
```

```
#>         Z
#> 0.5366599
```

$$\widehat{\beta}^{\text{IV}} = \frac{\widehat{\pi}}{\widehat{\gamma}} = \frac{1.138}{0.537} = 2.121$$

Standard errors

However, recall that our second-stage standard errors are not correct.

Second-stage results

```
#>          term estimate std.error statistic      p.value
#> 1 (Intercept) 5.637525 5.0620048  1.113694 2.681345e-01
#> 2       D_hat 2.121242 0.5180762  4.094460 8.707982e-05
```

2SLS results

```
#>          term estimate std.error statistic      p.value
#> 1 (Intercept) 5.637525 1.8003223  3.131398 2.293598e-03
#> 2           D 2.121242 0.1842559 11.512480 6.577085e-20
```

IV and 2SLS

Conclusions

1. IV/2SLS focus on **isolating some exogenous variation** in D_i via Z_i.
2. Important **requirements**: strong first stage, excludability, monotonicity.
3. IV and 2SLS **rescale the reduced form** with the first stage.
4. Estimates are **LATE from compliers**.
5. Different instruments can produce **different LATEs**.
6. A **weak first stage** can lead to problems.